Pokhara University
Faculty of Management Studies

Course Code: CMP 277
Course Title: **Data Communication and Networks**
Nature of the course: Theory + Practical
Year: Second,
Level: Bachelor
Semester: IV

Full marks: 100
Pass marks: 45
Credit Hrs: 3
Total periods: 48 hours
Program: BCSIT

1. Course Description

Data Communication and Networks is an introductory course designed to provide students with a comprehensive understanding of the fundamental concepts, principles, and technologies underlying data communication and computer networks. The course covers various topics, including digital communication system, network architecture, network reference models, standards and protocols, transmission media, transmission impairment, networking devices, network protocols, flow control and error control, congestion control mechanism and network security. Through a combination of theoretical knowledge and practical exercises, students will gain the necessary skills to design, implement, and manage computer networks effectively. It includes the following information about the course:

- Understand the basic concepts and terminology related to data communication and networking.
- Learn about the Nyquist Theorem and Shannon Channel Capacity Theorem.
- Explore the OSI (Open Systems Interconnection) and TCP/IP (Transmission Control Protocol/Internet Protocol) reference models and their layers.
- Analyze different network topologies, such as bus, star, ring, mesh, and hybrid, and understand their advantages and limitations.
- Learn about various networking devices, including routers, switches, hubs, and access points, and their roles in network communication.
- Examine different types of transmission media, such as twisted pair, coaxial cable, fiber-optic cable, and wireless, and understand their characteristics and applications.
- Explore networking protocols, including Ethernet, IP, TCP, UDP, HTTP, DNS, and SMTP, and understand their functions and mechanisms.
- Learn about network addressing and subnetting, including IPv4 and IPv6 addressing schemes.
- Understand the principles of error detection and correction, including parity check, checksum, and cyclic redundancy check (CRC).
- Understand the different types of interior and exterior routing protocols.
- Get the hands-on experience on connecting ethernet and RJ45 connectors and transfer the data.
- Study various network security concepts and mechanisms, including encryption, authentication, firewalls, intrusion detection/prevention systems (IDS/IPS), and virtual private networks (VPNs).

- Gain practical experience in configuring and managing computer networks using network simulation tools (cisco packet tracer and Wireshark) and real-world networking equipment.
- Gain practical experience on configuring different routing protocols on cisco devices.

2. General Objectives

After the completion of this course student will be:

- Understand the basics of data communication, networking, internet and their importance.
- Understand the capacity of noisy and noiseless communication channel.
- Analyze the services and features of various protocol layers in data networks.
- Differentiate wired and wireless computer networks.
- Analyze TCP/IP and their protocols.
- Recognize the different internet devices and their functions.
- Identify the basic security threats of the network.
- Design and configure peer-to-peer networks to share resources.
- Analyze requirements and design network architecture for a given scenario.
- Design and configure IP addressing schemes for a given scenario.
- Design and configure a client-server network and required network services for a given scenario.
- Evaluate and critique a design for a systems and network solution

3. Method of Instructions

General Instructional Technique: Lecture, Discussion, Readings, Question Answer
Specific Instructional Technique: Practical works, Project Based Learning, Self-Directed Learning, Industry Insights and Case Study

4. Course Contents

Specific Objectives	Contents
• Understand the concept of data and its significance in communication systems. • Explore different data transmission modes such as simplex, half-duplex, and full-duplex. • Identify the fundamental characteristics of data communication systems. • Illustrate the block diagram of a digital communication system. • Compare and contrast the advantages of digital transmission over analog transmission. • Explain the statement of the Nyquist Sampling Theorem • Apply Shannon's Channel Capacity Theorem to determine the maximum rate of error-free data transmission over	**Unit I: Introduction to Data Communication [5 Hrs.]** 1.1 Introduction to data communication 1.2 Data Transmission Modes (simplex, half-duplex, and full-duplex) 1.3 Fundamental Characteristics of Data Communication 1.4 Components of Data Communication System 1.5 Block Diagram of Digital Communication System 1.6 Introduction to Analog to Digital System 1.7 Advantages of digital transmission over Analog 1.8 Sampling Theorem/Nyquist Sampling Theorem 1.9 Numerical 1.10 Shannon Channel Capacity

a communication channel. ● Solve numerical problems to calculate channel capacity based on given parameters. ● Explain the concept of line coding and its role in digital data transmission. ● Discuss the characteristics and implementation of unipolar, polar, and bi-polar line coding schemes.	Theorem 1.11 Bit Rate Vs Baud Rate 1.12 Numerical 1.13 Line Coding (Unipolar, Polar and Bi-Polar)
Specific Objectives	**Contents**
● Define computer networks and explore their various uses and benefits in modern computing environments. ● Identify and describe different network topologies including bus, star, ring, mesh, and hybrid. ● Classify computer networks based on their geographical scope ● Differentiate between client/server and peer-to-peer networking architectures, highlighting their respective characteristics and applications. ● Define Protocols and Standards ● Explain the OSI (Open Systems Interconnection) Reference Model and its seven layers, illustrating the functions of each layer in the network communication process and compare it with TCP/IP ● Identify and describe various networking hardware components including Network Interface Card (NIC), Hub, Repeater, Switch, Bridge, and Router, elucidating their functions and roles in network communication. ● Familiarize with basic networking commands used for network configuration	**Unit 2: Introduction to Computer Networks[5 Hrs.]** 2.1 Definitions, Uses, Benefits 2.2 Network Topologies and its types (Bus, Star, Ring, Mesh, Hybrid,…) 2.3 Types of Computer Networks (PAN, LAN, MAN, WAN, CAN,…) 2.4 Networking Types (Client/Server, P2P) 2.5 Overview of Protocols and Standards 2.6 OSI Reference Model 2.7 TCP/IP Models and its Comparison with OSI 2.8 Networking Hardware: NIC, Hub, Repeater, Switch, Bridge, Router) 2.9 Basic Networking Commands
Specific Objectives	**Contents**
● Explore the duties and functions performed by the physical layer in transmitting data. ● Differentiate between guided and unguided transmission media. ● Compare and contrast Ethernet, Fast Ethernet, and Gigabit Ethernet	**Unit 3: Physical Layer and its Design Issues [7 Hrs.]** 3.1 Introduction, Design Issues and Duties of Physical Layer 3.2 Transmission Media: Guided (Twisted Pair, Coaxial Cable, Optical Fiber), Unguided (Radio Wave, Microwave,

technologies, including their data transfer speeds and applications. ● Introduce IEEE 802.11 (Wi-Fi) and IEEE 802.15.1 (Bluetooth) standards for wireless communication, including their specifications and applications. ● Identify and analyze common types of transmission impairments such as attenuation, distortion, and noise. ● Explain the concept of multiplexing and its importance in communication systems. ● Define different switching techniques including circuit switching, message switching, packet switching, and virtual circuit switching ● Define and quantify network performance metrics including bandwidth, throughput, latency, bandwidth-delay product, and jitter.	Infrared and Satellite) 3.3 Ethernet Cable Standards 3.4 Ethernet, Fast-Ethernet and Giga-Ethernet 3.5 IEEE 802.11 (Wi-Fi) and IEEE 802.15.1(Bluetooth) Standards 3.6 Signals 3.7 Transmission Impairment: Attenuation, Distortion and Noise 3.8 Multiplexing: TDM, FDM and WDM 3.9 Switching: Circuit Switching, Message Switching, Packet Switching, Virtual Circuit Switching 3.10 Network Performance: Bandwidth, Throughput, Latency, Bandwidth-Delay Product, Jitter
Specific Objectives	**Contents**
● Define the Data Link Layer and its role in providing reliable data transfer over the physical layer. ● Differentiate between Logical Link Control (LLC) and Media Access Control (MAC) sublayers. ● Discuss different framing techniques, including character stuffing, bit stuffing, and frame check sequence. ● Explain flow control mechanisms such as Simple Stop and Wait ARQ, Sliding Window, Go-Back-N ARQ, and Selective Repeat ARQ. ● Describe error detection techniques, including Parity Check, Checksum, Cyclic Redundancy Check (CRC), and Hamming Code. ● Discuss Random Access, ALOHA, Pure ALOHA, and Slotted ALOHA algorithms for channel allocation. ● Define VLAN and its role in logical segmentation of a LAN.	**Unit 4: Data Link Layer [8 Hrs.]** 4.1 Functions of Data Link Layer (DLL) 4.2 Overview of Logical Link Control (LLC) and Media Access Control (MAC) 4.3 Framing and its Types 4.4 Flow Control Mechanism: Simple Stop and Wait ARQ, Sliding Window, Go-Back-N ARQ, Selective Repeat ARQ, 4.5 Error Detection and Correction Techniques: Parity Check, Checksum, Cyclic Redundancy Check (CRC) and Hamming Code 4.6 Numerical 4.7 Channel Allocation Techniques (Multiple Access Techniques): Random Access, ALOHA, Pure ALOHA, Slotted ALOHA, 4.8 Carrier Sense Multiple Access (CSMA): CSMA/CD and CSMA/CA 4.9 VLAN
Specific Objectives	**Contents**
● Understand the role and functionality of the network layer in communication protocols.	**Unit 5: Network Layer [10 Hrs.]** 5.1 Network Layer and its Functions 5.2 IPV4 Address and its Headers, IPv4

● Explore IPV4 addressing and its header structure, including the classification of IPV4 addresses into different classes. ● Study various IPV4 addressing schemes such as static, dynamic, and automatic configuration. ● Differentiate between private and public IP addresses and their significance in networking. ● Learn the concepts of subnetting, subnet masks, Fixed-Length Subnet Masking (FLSM), and Variable-Length Subnet Masking (VLSM). ● Engage in numerical exercises to reinforce understanding of IPV4 addressing, subnetting, and related concepts. ● Explore the limitations and challenges associated with IPV4 addressing scheme. ● Introduce IPV6 addressing and its header structure as the next generation of internet protocol. ● Understand the strategies and challenges involved in transitioning from IPV4 to IPV6(Dual Stack, Tunneling and Header Translation) ● Study routing algorithms including adaptive and non-adaptive, as well as distance vector and link state routing. ● Explore popular routing protocols including Routing Information Protocol (RIP), Open Shortest Path First (OSPF), Enhanced Interior Gateway Routing Protocol (EIGRP), and Border Gateway Protocol (BGP). ● Understand tools and protocols used for network troubleshooting such as Ping, Internet Control Message Protocol (ICMP), Network Address Translation (NAT)	Classes 5.3 Private IP vs Public IP 5.4 Subnetting, Subnet Mask, FLSM, VLSM 5.5 Case Study 5.6 Issues with IPv4 5.7 Overview of IPv6 and its header format 5.8 Transition from IPv4 to IPv6 5.9 Routing: Adaptive and Non-Adaptive Routing, Distance Vector and Link State Routing 5.10 Famous Routing Protocols: RIP, OSPF, EIGRP, BGP 5.11 Ping(Lab Related), ICMP and NATing
. **Specific Objectives**	**Contents**
● Understand the role and functions of transport layer in the OSI Model ● Compare and contrast TCP and UDP. ● Understand SCTP (Stream Control	**Unit 6: Transport Layer [5 Hrs.]** **6.1** Introduction, Functions and Services **6.2** Transport Layer Protocols: TCP, UDP, SCTP, RTP and Their Comparisons,

Transmission Protocol) and Real-time Transport Protocol (RTP). ● Differentiate between connection oriented and connectionless services provided by the transport layer ● Explain the concept of port addressing and its significance and its significance in network communication. ● Understand the concept of ports and Sockets. ● Define congestion control and its importance in network performance. ● Compare open loop and closed loop congestion control algorithms ● Introduce traffic shaping algorithms such as Leaky Bucket and Token Bucket ● Describe queuing techniques used in scheduling data transmission.	TCP/UDP Segment Structure **6.3** Connection Oriented vs Connection less Services **6.4** Port Addressing Overview **6.5** Introduction to Ports and Sockets, Socket Programming **6.6** Congestion Control: Open Loop and Closed Loop Congestion Control **6.7** Traffic Shaping Algorithms: Leaky Bucket and Token Bucket **6.8** Queuing Techniques for Scheduling

Specific Objectives	**Contents**
● Define the concept and functions of application layers in network architecture. ● Explain the Hypertext Transfer Protocol (HTTP) and its significance in web communication. ● Discuss the secure variant, HTTPS, and its role in ensuring secure data transmission over the web. ● Define the Domain Name System (DNS) and its importance in translating domain names to IP addresses. ● Explore the working principle of DNS, including the hierarchical structure of domain names and the resolution process. ● Introduce email protocols, including Simple Mail Transfer Protocol (SMTP), Internet Message Access Protocol (IMAP), and Post Office Protocol (POP3). ● Compare and contrast FTP and SFTP in terms of security, authentication, and data transfer mechanisms. ● Understand the working principle of DHCP, including the DHCP discovery, offer, request, and acknowledgment	**Unit 7: Application Layers [4 Hrs.]** 7.1 Introduction and Functions 7.2 Web and HTTP/HTTPs 7.3 DNS and its Working Principle, DNS Query 7.4 Email Protocols: SMTP, IMAP, POP3 7.5 File Transfer Protocols: FTP, SFTP 7.6 DHCP and its working Principle

process.	
Specific Objectives	**Contents**
● Explore the objectives of network management, including monitoring, configuration, troubleshooting, and optimization of network resources. ● Introduce the Simple Network Management Protocol (SNMP) and its role in network management. ● Explain the data security, its importance and types ● Define network security and its importance in protecting data, systems, and network infrastructure from unauthorized access, misuse, and attacks. ● Introduce countermeasures and best practices to mitigate network security risks and prevent network attacks. ● Define symmetric key cryptography and asymmetric key cryptography and explain their respective principles of operation. ● Explain the concepts of tunneling and encryption in VPNs and their importance in securing remote connections.	**Unit 8: Network Management and Network Security [4 Hrs.]** 8.1 Introduction to Network Management 8.2 SNMP Protocol 8.3 Data security, its types, key risks in data security 8.4 Network Security: Introduction, Types of Network Attacks, Countermeasures to prevent network attacks 8.5 Cryptography: Symmetric Key and Asymmetric key Cryptography 8.6 VPN, Firewalls and Proxy

5. Laboratory Work

i. Build and Test a Network Cable (Ethernet Cable and RJ45 Crimping)
ii. OS (Ubuntu/Windows) Installation, Practice Basic Networking Commands (ipconfig, ipconfig/all, ping, arp, netstat, nbtstat, nslookup,….)
iii. File sharing between PCs connected across static IP addresses
iv. Introduction to Cisco Packet Tracer, Connection of Simple Network and Ping all the devices
v. VLAN Creation and VLAN Trunking cisco packet tracer/cisco devices
vi. Basic Router Configuration, Static Routing Implementation
vii. Subnetting Implementation cisco packet tracer/cisco devices
viii. Implementation of Cyclic Redundancy Check using any programing language
ix. DHCP Implementation cisco packet tracer/cisco devices
x. Implementation of Dynamic/interior/Exterior Routing Protocols (RIP, OSPF, EIGRP and BGP) cisco packet tracer/GNS3/cisco devices
xi. Firewall Implementation, Router Access Control List (ACL) cisco packet tracer
xii. Packet Capture and Header Analysis by Wireshark
xiii. DNS, Web, FTP, DHCP Server Configuration using cisco packet tracer/cisco devices
xiv. Case Study: Visit Nearby Enterprise Network or ISP or Nepal Telecom BTS/BSC/MSC/Data Centers and Prepare report.

xv. Lab Exam, Report and VIVA

6. Evaluation system and Student's Responsibility

In addition to the formal exam(s), the internal evaluation of a student may consist of quizzes, assignments, lab reports, projects, class participation, etc. The tabular presentation of the internal evaluation is as follows.

Internal Evaluation	**Weight**	**Marks**	**External Evaluation**	**Marks**
Theory		**30**	**Semester End**	**50**
Attendance & Class Participation	10%	3		
Assignments	20%	6		
Presentations/Quizzes	10%	3		
Internal Assessment	60%	18		
Practical		**20**		
Attendance & Class Participation	10%	2		
Lab Report/Project Report	20%	4		
Practical Exam/Project Work	40%	8		
Viva	30%	6		
Total Internal		**50**		
Full Marks: 50 + 50 = 100				

Student's Requirements

Each student must secure at least 45% marks separately in both internal assessment and practical evaluation with a minimum of 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such score will be given NOT QUALIFIED (NQ) to appear the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. ***Students are required to complete all the requirements defined for the completion of the course.***

7. List of Tutorials

- **Unit I –** Nyquist Sampling Theorem, Nyquist Rate, Shannon Channel Capacity, Bit Rate vs Baud Rate and Line Coding
- **Unit IV-** Error Correction and Detection Techniques: Checksum, Cyclic Redundancy Check (CRC), Hamming Code
- **Unit V-** IPV4 Subnetting

8. Prescribed Books and References

Prescribed Text Books

1. Data Communications and Networking, 4th Edition, Behrouz A Forouzan. McGraw-Hill
2. Computer Networking; A Top Down Approach Featuring The Internet, 2nd Edition, Kurose James F., Ross W. Keith PEARSON EDUCATION ASIA

3. Peterson and Davie, "Computer Networks, A Systems Approach", 5th Edition, Morgan Kaufman, ISBN: 978-0123850591.

Reference Books

- Steven Holzner,"HTML Black Book" Dremtech press.
- Web Technologies, Black Book, dreamtech Press
- Web Applications : Concepts and Real World Design, Knuckles, Wiley-India
- Internet and World Wide Web How to program, P.J. Deitel & H.M. Deitel. Pearson.
- Computer Networks, a top down approach, 6th Edition, Kurose/Ross.